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Reconstructing Combinatorial Inventions Through Design Problem Analysis

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Abstract

Combinatorial invention—the process by which new technologies emerge through the combination of existing ones—is central to technological evolution, yet it remains poorly understood. This paper develops a novel framework for analysing combinatorial invention by recognising its dual nature as both a design problem-solving process and a lineage-generating process. While existing approaches often treat combinatorial invention as a stochastic process mapping precursor technologies to composite ones, this framework reveals how the challenge of interfacing components generates multiple concurrent technological traditions. By identifying genuine design sub-problems—those requiring solutions that were novel, necessary, and culturally transmitted—we can reconstruct aspects of combinatorial invention processes through the phylogenetic traces their solutions left in the archaeological record. Using the invention of hafted tools by Neanderthals and their African contemporaries as an illustrative case study, I show how this framework reveals a complex, multi-tiered invention process involving functional organization, material implementation, and assembly procedures. The combinatorial invention of complex technologies should be understood not merely as producing composite technologies, but in addition as one generating a manifold of technological lineages through the solving of multiple design sub-problems, specifically traditions of interfaces between the components of the technologies. The proposed framework provides preliminary methodological grounds for reconstructing prehistoric invention processes.

Keywords: invention, combinatorial invention, problem-solving, design problems, technological evolution, combinatorial evolution, hafted tools, pre-*sapiens* technology

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1. Introduction

Combinatorial invention—sometimes called 'recombination'—is a central process of technological evolution (Arthur 2009; Charbonneau 2016; Mesoudi and Thornton 2018). Nearly all human technologies have emerged through the combination of existing knowledge, techniques, materials, and artefacts. Ford's first automobile, the quadricycle, integrated an internal combustion engine, a chain attached to two gears, four bicycle wired wheels, a boat tiller, a padded seat, and other components. Even the earliest technologies invented by Modern humans follow this pattern. For instance, stone-tipped harpoons and arrows are composites combining points and microblades with shafts using adhesives and bindings, all pre-existing technological elements.

Combinatorial invention is not a prerogative of our species. While the capacity to combine technologies has not yet been observed in any meaningful way in non-human animals, there is little controversy regarding its existence in Neanderthals and their African contemporaries. Indeed, the oldest known case of a technological novelty invented through the combination of pre-existing components is that of the hafted tool (Barham 2013; Kuhn 2020; Wilkins et al. 2012; Wragg Sykes 2020): about 500kya, some pre-*sapiens* hominins had the brilliant idea to combine their hand-held stone tools to a shaft that would serve as a handle, an invention so important in its ramifications that Barham (2013) references it as initiating the first industrial revolution.

Despite its importance, there is surprisingly little work that examines just how combinatorial invention works. The most elaborate study on this issue is that of Arthur (2009), but his analysis is set specifically in the context of recent techno-scientifically informed, institutionally supported, professionalised engineering practices, and therefore fails to cover most of human (pre)history of inventions. More recently, Charbonneau (2016) developed a preliminary understanding of technological combination, focusing specifically on how techniques—defined as "hierarchically organized set[s] of actions and decisions leading to the satisfaction of a specific, intended goal" (p.378)—can be decomposed into modules and recombined into novel technological traditions. Here I pursue this project by focusing on the sorts of processes involved in creating new technologies through combinatorial invention.

While combinatorial invention itself remains understudied, combinatorial *evolution* has been analysed through mathematical and computational models (Enquist et al. 2011; Kolodny et al. 2015; 2016; Koppl et al. 2021; Weitzman 1998), and experiments (Derex and Boyd 2015; 2016;

MacCallum et al. 2012). These studies examine how populations of individuals capable of generating combinatorial inventions lead to technological change and diversification (e.g., Kolodny, Creanza, and Feldman 2015; Koppl et al. 2021). Some studies focus on comparisons with other forms of invention processes (e.g., Lewis and Laland 2012; Enquist et al. 2011) and demographic factors such as population size and connectivity (e.g., Creanza et al. 2017; Derex et al. 2018; Derex and Boyd 2016).

In these studies, combinatorial invention is treated as an event whose internal mechanics can be black-boxed and abstracted away without loss of explanatory validity. This is done by modelling combinatorial invention as a stochastic process where existing technologies randomly combine in a lock-and-key manner (for an exception, see Arthur and Polak [2006]). For instance, in some computational models, combinatorial invention is reduced to the random sampling of precursor elements leading to the emergence of a novel combination and, for some studies, the resulting combination exhibits degrees of efficiency (or fitness) different from those of the combined precursors. Detailed here are the input conditions and how they map to output results, but the process of combination effecting this mapping is not examined. In other words, how precursor elements fit together, or how they are made to fit one another, is left unexplained.

What these models overlook is that existing technologies rarely combine seamlessly nor randomly. Not any precursor technologies can be usefully combined with any other, and when they can, they typically require carefully designing interfaces between them to ensure they can function together as intended (Arthur 2009; Charbonneau 2016). Moreover, the necessity for inventing interfaces that yield working composite technologies itself generates multiple new traditions beyond that of the composite technology: the interfaces themselves often form technological lineages of their own and new procedures to assemble the composite technology must be designed. For instance, inventing the hafted spear does not merely consist of putting a point at the tip of a shaft. In addition, one must design a hafting system so that the movements and forces applied to the handle are transferred to the point. Hafting systems must be transmitted across generations as well, and there are various traditions of such systems (Barham 2013; Rots 2010; see below). A theory of combinatorial invention must therefore explain not just which precursors are joined together (how they are selected), but also how technologies can be functionally and materially interfaced, and how this process generates multiple sub-inventions which form traditions of their own. Models that ignore these aspects risk either approximating evolutionary dynamics with unverified assumptions or fundamentally mischaracterising the combinatorial evolutionary process.

In contrast to the modelling literature, here I develop an understanding of combinatorial invention as a problem-solving process generating multiple new technological lineages. Using the invention of hafted tools by Neanderthals and their African contemporaries as a case study, I propose a method that decomposes combinatorial invention into distinct problem-solving layers or tiers, each composed of several design sub-problems. I propose that by tracking design sub-problems involved in the larger invention of a composite technology, we can open the black box of combinatorial invention by treating it as a process rather than as a chance event and provide a deeper understanding of technological evolution.

For reasons of space, this paper does not directly challenge the hypothesis that combinatorial invention emerges primarily through chance-based, serendipitous events rather than complex cognitive mechanisms, i.e., the view that combinatorial invention is a process that "can create complex cultural adaptations without any intelligence, save that required to distinguish among more- and less-successful teachers" (Boyd et al. 2011, 10921). However, by reconceptualising combinatorial invention as a design problem-solving process generating new lineages, the framework developed here provides new grounds for evaluating competing explanatory models, from those emphasising chance-based discovery to those proposing the involvement of more complex cognitive capacities.

2. Terminology

Given the paucity of studies examining combinatorial invention, there is little, if any, order in the terminology used when discussing such process. To maximise clarity, some terminology and conceptual analysis are required.

Combination as a concept refers to the putting together of two or more items, forming through their joining a novel item. In the context of technological evolution, those items that are put together are referred to here as 'precursor' technologies, underlining their pre-existence to the composite result into which they are combined. When referring to these precursors as parts of the composite result, I use the term 'component' instead. Since 'invention' is a term that is both a substantive and an activity, invention as an *end-result* will be referred to as a composite invention (e.g., the hafted spear), whereas the *process* of joining together precursors into such a composite invention is referred to as the process of combinatorial invention (e.g., the process by which the hafted spear was invented).

For a technology to be a *composite*, it must satisfy two conditions: (1) its components must be organised in a functional part-whole relationship, and (2) these components must, at least in

part, be precursors repurposed into the composite, forming a lineage relation between them and the composite invention.

Regarding the first condition, a composite is a complex technology that is made of parts that are organised in such a way that the causal interactions (e.g., mechanical interactions) between the components yield the composite technology. This arrangement means the composite technology operates as a functional complex whole exhibiting capabilities beyond the mere sum of its components' individual causal contributions. The organisation of components is inherently hierarchical, with lower-level interactions between the components giving rise to the higher-level functioning of the composite (Arthur 2009). Therefore, not all instances of bringing objects together qualify as true technological composites. For example, joining two haystacks creates a larger pile but introduces no new functional properties or interactions between the original components. Such purely additive arrangements lack the essential causal integration that characterises technological composites.

Regarding the second condition, a new technological composite involves more than the mere temporal precedence of its components: a new composite establishes a lineage with its precursor technologies as these are repurposed as components of the composite. This reuse of established technologies as constituent parts creates a line of descent between the composite and its technological predecessors. What makes combinatorial invention special is precisely this exploitation of existing technologies as building blocks, creating new technological lineages by joining existing ones. It is also possible that some but not all components of pre-existing composites are used as precursors to form a new composite technology (for instance, Ford used bicycle wheels but not full bicycles to invent his quadricycle). In such cases, we can speak of the composite as a *recombination*.

Combinatorial invention refers to the process by which an individual (or group of individuals) effectively exploits precursor technologies to produce new technological composites. This process includes the range of activities and resources by which composite inventions are produced out of existing technologies. In contrast, *combinatorial evolution* refers to the population-level effects and evolutionary dynamics resulting from individuals capable of combinatorial invention.

In this paper specifically, I understand *technology* to encompass the range of knowledge, techniques, and artefacts used for practical activities and that are learned socially. I also include manufactured or extracted materials as technological products (e.g., refined ore) when

obtaining such materials requires some form of social learning or cultural transmission (e.g., learning from an expert how to smelt ores).

3. Invention as an evolutionary process

Contrary to the long-standing myth that inventions are *sui generis* products of genius minds, from a cultural evolutionary perspective, technologies are the result of an intergenerational cumulative process of invention and transmission (Mesoudi and Thornton 2018; Tomasello 1999). In cumulative cultural (or technological) evolution, invention is the process by which something new is created out of something old (Basalla 1988; Charbonneau 2015; Henrich 2010). It is the process by which new technological variants are generated by modifying or repurposing existing ones[1]. Complex technologies, therefore, are rarely, if ever, invented at any one moment by a single individual, but instead develop through time as generations of individuals add novel modifications to existing technological lineages (Basalla 1988; Boyd and Richerson 1996). In this sense, complex technologies are invented over multiple generations through a culturally distributed process (Muthukrishna and Henrich 2016).

In this section, I argue that there are two sides to invention (and therefore combinatorial invention). First, invention is a design problem-solving process where individuals use existing technologies to create novel technological solutions. Combinatorial invention relies on the causal processes and activities by which individuals manage to use and interface precursor technologies to serve as components of novel technological composites. Approaching combinatorial invention as an individual-level problem-solving process contrasts with the random lock-and-key simplification used in studies of combinatorial evolution by focusing on the activities of invention rather than relegating them to random luck.

Second, invention is also a lineage-generating process where new technologies, such as composite technologies, are derived from and built upon existing ones. Invention therefore produces a form of intergenerational continuity between technological variants at the population level, initiating novel traditions by deriving them from existing ones. This dual nature of invention, rarely made explicit in the literature (but see Charbonneau 2015), provides a

[1] Some inventions are made by individuals interacting directly with their environment without relying on existing technologies to solve the invention. Such 'inventions from scratch' (Charbonneau 2015, 201) do not form lineage relations between existing and novel variants and so will not be considered here, as they are generally limited to the very onset of technological evolution (but see Haidle and Schlaudt 2020). In contrast, I am concerned here with inventions that can be understood as modifications of existing technologies, i.e., those that depend on existing technological traditions to be solved (see Charbonneau [2015] for a detailed discussion of the difference between inventions from scratch and modifications).

framework for understanding how invention operates both at the individual level, through problem-solving activities, and at the population level, through the formation of new technological traditions, or lineages, derived from older ones.

3.1 Invention as design problem-solving

Invention provides new solutions to problems, be they old problems in need of better solutions or new problems yet unsolved. A problem—whether social, ecological, mathematical, or otherwise—emerges when individuals or groups of individuals face a situation defined by (1) specific starting conditions, (2) limited resources and means to use these resources, and (3) unfulfilled needs or tasks (Newell and Simon 1972). As a process, problem-solving thus involves determining how to address some task using available means within the given constraints of the starting conditions. For instance, a hungry tribe (starting condition) can capture and kill prey (the task) by using existing tools (available means) to dig a hole and floor it with sharp-pointed sticks (the solution).

Inventing is a special problem-solving activity in that it solves *design* problems, i.e., the solutions it provides are novel means that can later be used, by the inventor or someone else, to solve other problems (Charbonneau 2025). Inventing therefore consists of using available means (the old) to design and produce additional means (the new). Combinatorial invention is no exception: it provides a new means—the composite invention—by using precursor technologies to design and implement the composite invention—the solution to the design problem. Explaining how an invention process works thereby consists of explaining the activities engaged in the design problem-solving process. Approaching invention as problem-solving provides a causal understanding of how novelties can emerge out of existing technologies.

As pointed out by Arthur (2009), invention is typically a recursive and prolonged process. Complex design problems rarely have straightforward solutions. Instead, they require solving multiple intermediate design problems that build upon each other before the main design problem can be addressed. For instance, to make copper jewellery, one needs to smelt copper ore, which in turn requires one to find a way to produce high temperatures. One solution is to enclose fuel in a closed space to retain maximum heat, for example by digging an earth oven. These intermediate design problems can then lead to additional layers of sub-problems, each of which may in turn require the solving of additional intermediate problems. Following on the previous example, as an enclosed space, earth ovens can easily asphyxiate the fire, which requires a means to provide oxygen. One solution was to invent ceramic blowpipes, which in

turn can help raise the temperature of the oven (Radivojević and Roberts 2021). This nested structure means that a single invention may comprise several sub-inventions. Since each sub-invention represents its own discrete design problem, each may be solved by different individuals, in different locations, and at different times (more on this below).

As cumulative technological evolution is an intergenerational and gradual process, complex technologies have no clear moment of inception. From an evolutionary perspective, it is crucial to understand that the boundaries of what constitutes the evolutionary history of 'an invention' are largely determined by the investigator's analytical choices. These choices involve two key decisions: first, one identifies which technological novelty is of interest as the end result of an invention process, and second, one determines which technological and cultural context will serve as the starting conditions for the analysis. These choices shape which design problems are considered part of the invention process under study. For instance, when studying the invention of metallurgy, an investigator must decide what it means to have invented metallurgy—e.g., by using cold or hot extraction techniques—and what precursor context to start from—e.g., with the early use of minerals as decorative pigments or with the first traces of slagged masses (Radivojević 2015). Different choices about the boundaries of the invention process will lead to different analyses of which sub-problems need to be solved. This does not make the analysis arbitrary, however, as the actual steps of the process are not arbitrarily determined. Rather, it requires being explicit about the methodological choices concerning the range of steps that are of interest and just what novelty one aims to study the origins of.

3.2 Invention as a lineage-making process

While understanding invention as a recursive design problem-solving process is well established, its role as a lineage-making process is often overlooked. Within the fields of cultural evolution and evolutionary archaeology, lineage formation has been attributed primarily to cultural transmission—the process by which technological knowledge and techniques are passed between individuals and across generations (Richerson and Boyd 2005). Cultural transmission functions as an inheritance system, ensuring both the temporal continuity of technological traditions and their diffusion (Boyd and Richerson 1985; Cavalli-Sforza and Feldman 1981). Invention, by contrast, has been mostly understood as a variation-generating process that disrupts existing lineages rather than contributing to their formation (e.g., as a process of individual learning that reduces transmission fidelity [Boyd and Richerson 2005]).

While invention does break away from the past by producing novel traits into a population's technological repertoire, it simultaneously creates genealogical connections between precursor and novel technologies. Genealogical connections arise through invention because new technologies are derived from and built upon existing cultural knowledge and technological precursors. Thus, even though an invention may originate through an individual's activity, it inherently establishes genealogical (phylogenetic) relationships between the precursor technologies from which the novelty was derived, thereby potentially initiating a lasting tradition or lineage. In this sense, invention serves as a lineage-making process.

Approaching invention as a form of design problem-solving provides a mechanistic link between individual-level invention processes and population-level lineage formation. When individuals engage in design problem-solving, they draw upon existing technological traditions as available means shaping their problem space. These precursor traditions serve dual roles: at the individual level, they constitute the available resources and means that shape the designing of novel solutions; at the population level, they represent the precursor technological traditions from which new technological lineages emerge. This correspondence means that successful solutions to design problems simultaneously generate both a novel technology at the individual level and a potential new technological tradition at the population level.

Approaching invention this way also provides a means to study invention as a process spanning multiple timescales. One can choose to study invention over a short timescale, such as when an individual living in a specific cultural background, one populated with different techniques, materials, and so on, can, using these resources (the starting conditions and available means for the invention) solve a design problem (or series of problems) to produce a novel technology. If the new technology is picked up by others and is transmitted within the population and across generations, it will have successfully formed a lasting tradition, with the invention process having both created the new technology and having phylogenetically linked it to its precursor form. In this situation, the generation of a new lineage and the invention process all occur within the lifespan of a single individual.

Other inventions, such as those of complex technologies, are best studied over longer, multi-generational timescales. Archaeologists studying invention typically do so (Frieman 2021). For instance, asking how metallurgy (Roberts and Radivojević 2015) or the wheel (Alacoque et al. 2024) were invented requires reconstructing a multi-generational, multi-invention process. As discussed above, invention is often a complex process, one requiring the solving of multiple design sub-problems, each of which in turn can also require more design sub-problem-solving.

This recursiveness means that solving a complex design problem often means providing novel solutions to several design sub-problems, each of which can be made by different individuals at different times and places, effectively providing an intergenerational form of division of labour (i.e., a cumulative process). Therefore, so long as the investigator identifies a context serving as a starting point and one as an end point for the invention process—using available evidence and filling-in gaps by judiciously hypothesising additional context that may have been lost—one can start reconstructing the multi-generational invention process by tracking the different sub-problems and their solutions (sub-inventions) that had to be solved along the way.

Finally, viewing invention as a multi-generational problem-solving process does not mean the inventors were consciously working towards the "final" complex technology or planning out the steps to create it. Long-term technological development can emerge from individuals solving immediate, concrete problems, with no long-term teleological plan or goal of discovery. To the extent that invention is teleological, it should concern proximate goals, i.e., goals conceivable by an individual and within reach during their lifetime (Charbonneau 2025). Over generations, these incremental improvements and inventions can cumulate into complex technologies that no single individual could have devised on their own. Finally, solving design sub-problems across people and time allows for a reduction in the cognitive needs for complex (combinatorial) inventions, as each sub-problem is less complex to solve than the higher-level design problem they jointly solve. This provides a more gradualistic, cognitively less complex story than suddenly inventing a fully-fleshed complex tool (see Section 9).

4. Reconstructing combinatorial invention

Approaching invention as a design problem-solving process that produces new lineages opens new ways to study invention within an evolutionary framework. While the method applies to invention broadly, here I focus on combinatorial invention specifically. The strategy provided here consists of reconstructing the range of design problems and sub-problems that were solved for a specific invention to come about[2]. As discussed above, the investigator must first determine what the invention of interest is and determine a starting point from which the invention is hypothesised to have emerged. The reconstruction then proceeds by identifying

[2] Several projects in archaeology have employed similar strategies, reconstructing the series of steps needed for the invention of some technology of interest. Work on the invention of metallurgy (Radivojević 2015; Roberts and Radivojević 2015), the invention of the wheeled vehicle (Alacoque et al. 2024), and of fire (Chazan 2017; Gowlett 2016) have employed similar strategies, often complementing them with additional sources of evidence, such as the use of computational models (e.g. Alacoque et al. 2024). So, while the reasoning behind the strategy I propose here is not exactly new, to my knowledge it is the first time that its epistemology has been explicitly formulated.

design sub-problems that need solving along the way, providing a series of sub-solutions (sub-inventions) and problem-solving activities involved in bridging the end result to the starting conditions.

The identification of design sub-problems and their solutions can proceed in several ways. One strategy is to simply provide a plausible causal story as to how the composite invention could have been invented. Here, the archaeologist (or historian) provides a narrative that allows working from the starting conditions to the end result through a series of problem-solving steps. However, such narratives bring two risks. First, there is always the risk that the causal story reflects the reasoning process of the investigator more than the effective process by relying on anachronistic modes of reasoning, such as hindsight, that were not available to the inventors themselves. In turn, this can introduce unwanted teleology within the causal story, for instance, by identifying the shortest series of sub-solutions that could lead to the composite technology. While such direct pathways may be possible, longer routes and detours could also have been taken, reflecting local concerns of the inventors rather than a guided, long-term R&D process that would benefit only later generations (see Section 3.2). Second, there are likely multiple pathways by which the same invention can come about. For instance, wheeled vehicles could have been invented through the transfer and adaptation to transportation purposes of the fibre spinning wheel (Yashuv and Grosman 2024), of children's toys (Bondár 2012), or through the design of wheeled mining carts (Alacoque et al. 2024). While archaeological findings can help discredit and reject certain narratives, there remains the risk that we end up providing just-so stories of how an invention may have come about, with little other than the investigator's intuitions to prefer one story over another.

The alternative proposed here is that, instead of reconstructing from the starting conditions a causal story of inventions and sub-inventions, one instead works back from the complex solution by looking at the phylogenetic traces that the design sub-problems for that invention left behind through their solutions. This means that it is not just the full artifacts that form lineages, but also their component parts, i.e., complex artifacts are modular systems the modules of which can form their own lineages (Charbonneau 2016). As discussed above, complex inventions can be the results of several sub-inventions, each answering design sub-problems of the larger design problem-solving process. This means that the sub-solutions are themselves part of the larger invention. These sub-inventions can either become integral components of the composite invention (e.g., a new component was invented in the combinatorial process) or serve as supportive traditions necessary for the composite technology to be successfully used, built and transmitted (e.g., a procedure to produce some

necessary component of the technological composite). In each case, the sub-solutions must themselves form traditions if they are to be transmitted along with the composite invention (Charbonneau 2016). Should we find phylogenetic traces of these sub-inventions persisting together with the invention of the composite technology, these would count as evidence that the design sub-problems they solved were part of the combinatorial invention process.

The strategy I propose, then, is to decompose the complex composite invention into the range of sub-solutions that serve as components or that serve as supportive solutions that themselves need to be transmitted for the composite invention to persist across generations. In order to reconstruct an invention process, we need means to determine which sub-problems were genuinely part of the invention process of the technology. Accordingly, I propose that the combinatorial invention of a composite technology be decomposed into design sub-problems by identifying sub-solutions that jointly satisfy three conditions: (1) novelty, (2) necessity, and (3) transmission.

Novelty—A sub-problem is genuine if it is novel to the context of the invention of the composite and therefore requires designing a novel solution to be solved. The rationale here is to avoid including sub-problems with solutions that predate the invention process. Design sub-problems that are already solved within the starting conditions defined by the investigator provide existing means that can be exploited during the invention process, but these are not generated during that process.

Necessity—A design sub-problem is genuine if it was necessary to provide a solution to that sub-problem in order to invent the composite. The rationale here is that design sub-problems and solutions that are unnecessary may or may not be part of the invention process, and therefore do not provide sufficient evidence of their role within the invention process. This condition can be relaxed, however, by accounting for alternative pathways by which the same invention could have been made. In such cases, the alternatives should be disjunctively necessary, i.e., at least one pathway would need to hold true for the invention to have happened at all. Not all sub-solutions will therefore be necessary individually, but without them or some range of alternatives, inventing the composite would not have been possible.

Transmission—A design sub-problem is genuine if its solution is transmitted culturally and is detected in the archaeological or historical record as a lasting tradition. A design sub-problem for which no evolutionary signal can be found should therefore be excluded hypothetically. However, not all lasting regularities involve transmission, as some aspects of a tradition can be reconstructed anew every generation without it being transmitted (Miton and DeDeo 2022). In

this sense, non-obvious solutions should be transmitted. A strong signal of transmission is the observation that the solution can vary in different populations synchronically or diachronically. Identifying variant solutions to the same sub-problem indicates that the sub-problem is genuine when the variation is best explained as different means to solve the same design problem, and can be observed to be adopted by different populations and/or at different times (i.e., it is cultural).

These three criteria help us identify the range of design sub-problems and sub-solutions forming the invention process of a novel composite technology. However, these do not necessarily yield all the design sub-problems and sub-inventions that were part of the invention. The framework proposed here does not pretend to be capable of identifying the whole process of invention, but only to provide plausible hypotheses in an evidence-based manner. Let us now turn to the combinatorial invention of the hafted tool to illustrate what the framework proposed here can do.

5. The invention of the hafted tool

The earliest known case of intentional toolmaking dates back to at least 3.3 million years (Harmand et al. 2015). For the three million years or so that followed, our ancestors were exclusively producing single-object implements serving multiple functions, such as scraping, cutting, clubbing, piercing, etc. All of these were objects made of a single material—be it stone, wood, bone, or else—which their user would manipulate directly with their fingers and palms. During the Middle Pleistocene, about 500kya, a revolution in toolmaking occurred: someone came up with the brilliant idea of designing a hafted tool, i.e., to add a handle to those hand-held implements, thereby producing tools with multiple parts and separating the hand of the user from directly manipulating the working end of the tool (Barham 2013; Kuhn 2020; Wragg Sykes 2020).

A hafted tool is composed of at least three functional parts—a working end, a handle, and a joint—each separate objects typically made of different materials (Figure 1). The working end directly interacts with some other materials by mechanically transforming it. The handle is held and oriented by the user to transfer specific movements and forces to the working end, while the joint binds the working end and the handle together, permitting both the transfer of mechanical work between the two other parts and ensuring they remain attached in the process.

[FIGURE 1]

The specific uses permitted by the hafted tool are determined by the nature of its working ends and the range of movements and forces the user can transfer to it by manipulating the handle. A hafted knife consists of a blade parallel to the long axis of the handle and is used to cut some material by having the user push down on the handle or move it in a sawing manner while pressing down. A hafted adze has a cutting working end oriented perpendicularly to the haft, which is manipulated by swinging the handle towards some material to cut through or scrape its surface. Etc.

The earliest known case of hafted tool technology is attributed to pre-*sapiens* hominins, such as the Neanderthals and their African contemporaries (Barham 2013; Wilkins et al. 2012). No full specimen has been found within the archaeological record, due to the use of perishable materials employed for the joint and haft (and plausibly for some working ends, such as those made of bones, etc.). Knowledge of the existence of these tools is instead derived from the combination of indirect evidence (Rots 2010), such as microwear and damage analysis (e.g., marks of hafting on the working end), the presence of hafting residue (e.g., the presence of tar and other adhesives on the working end), and the morphology of the remaining stone implements (e.g., micro-blades too small to be hand-held tools, or tang on points). Through these means, archaeologists have established that these pre-*sapiens* hominins invented a diversity of hafted tools, including hafted scrapers, spears, and knives (Barham 2013).

The hafted tool satisfies the two criteria for being a composite (see Section 2). Consider, for instance, the case of the stone-tipped hafted spear (Figure 2). First, the stone point, the wooden shaft and the binding materials (such as adhesives and twines) are physically organised to interact in such a way that they provide the hafted spear its distinctive function. This organisation also ensures that the physical organisation of the components provides the composite tool with capabilities beyond those of its individual parts: the shaft leverage amplifies the force and directs the movement to the point, while the binding materials ensure the transfer of the force and movements to the working end. This is not merely an additive assembly: the hafted spear's functioning depends on the specific arrangement and interactions between its components, each of which serves a distinctive causal role in the spear's overall functioning. Second, the hafted spear is also made of components that existed as independent technologies before their integration into the composite, thereby creating lineage relation with these precursors. Stone tools were already in use as hand-held cutting and piercing tools, and the shaft likely derived from simple wooden spears. A similar analysis could be provided to the other known hafted tools made by pre-*sapiens* hominins.

Hafted tools, therefore, are likely descended from precursor technologies now functionally organised as components in a functional part-whole system. While this allows us to hypothesise that the hafted spear is the result of a combinatorial invention, it does not explain how it was invented. In the next three sections, I provide a schematic reconstruction of the invention of the hafted tool along the lines described above. The reconstruction is meant first and foremost to illustrate how one can deploy the proposed framework. Therefore, I do not suggest that the hafted spear was, indeed, the first hafted tool. A deeper analysis and more contextualisation than was possible here are required.

I propose to organize the design sub-problems into three tiers, each of which speak to one another but each being addressable in a more or less independent measure: (1) the sub-problems and solutions related to the functional organisation of the composite tool (Section 6), (2) the sub-problems and solutions concerning the means by which the physical parts can be connected into a working tool (Section 7), and finally (3) the sub-problems and solutions regarding the specific procedures that must be followed if the composite is to be assembled into a working form (Section 8).

For this reconstruction, I focus on the invention of the hafted spear, supposing for the sake of the argument that it was the first hafted tool invented. Following Barham (2013), as a starting condition I assume a technological repertoire made of hand-held tools, the range of techniques and knowledge required to make, use and retouch those tools, together with the access to the materials to produce adhesives (vegetal or mineral) and binders (human and/or animal hair, twines, tendons, etc.). I also include the controlled use of fire. Importantly, I assume the existence of non-hafted (or simple) wooden spears in this starting repertoire, along the lines of those discovered in Schöningen (Thieme 1997; Schoch et al. 2015). While it is likely that binders (and possibly adhesives) were available before the invention of hafted tools, I will assume they were not so as to provide a more streamlined discussion (this assumption can be relaxed, with no real impact on the present aims). This analysis, then, provides a sketch of how the first hafted spear could have been invented out of this context by identifying the range of design sub-problems one needs to solve for its combinatorial invention.

A few caveats. For reasons of space, the analysis is preliminary, as it focuses only on a curated set of higher-level sub-problems involved in the invention of the hafted spear, i.e., those which were most helpful to demonstrate the complexity of the invention process and the necessity of inventing interface systems. It is also preliminary in that neither do I attempt to order the sub-problems chronologically—i.e., determine which sub-problems were solved before others—nor

do I aim to identify the range of causal mechanisms, such as cognitive processes, that might have been employed for solving this complex constellation of design problems (see Discussion). The case study is illustrative, aiming to show the potential of the design problem analysis method rather than providing a strong case about the invention of the hafted spear specifically.

6. Inventing a functional organisation

An important aspect of designing a new technology consists of inventing a functional organisation whereby the different roles of the components can be interfaced so as to form a functioning technological composite (Arthur 2009). A composite tool's functional organisation consists of (1) its use-function—the intended use or the range of tasks for which the composite is meant to be used for—(2) the sub-functions of its components—which functions the parts serve within the functional economy of the whole, and (3) the specific functional organisation of the sub-components—how the components are organised in relation to one another such that they can bring about the use-function of the composite whole (Cummins 1975).

The hafted spear's functional organisation can be decomposed accordingly. As a hunting and/or defensive weapon, it's use-function is to injure or kill an animal by piercing its skin. This could be done by using the weapon through thrusting, as a projectile weapon, and/or as a receiving weapon (exploiting the moving animal's own momentum against itself). Each of its subcomponents serve specific sub-functions: the role of the point is to pierce through an animal's skin, the role of the shaft is to serve as a handle that can be manipulated to drive the point through an animal's skin, and the role of the haft, adhesives, and/or binders is to physically join the point to the shaft. Moreover, the components are organised so as to serve the use-function of the spear: the point should extrude outwards, opposite but along the axis of the shaft, so that aligning the shaft towards an animal can lead the point to pierce its skin. The joint must also ensure the shaft can transmit the movements and forces applied to it by the user to the point's piercing end (Figure 2).

[FIGURE 2]

The simple spear can be analysed in a similar way, but with some key differences, as it is not a composite. Indeed, there are no components to the simple spear as it is a single object. Instead, it has sections of its body serving different functions, located at different locations on the tool. So, while the use-function of the single-object and composite spear is the same—they both serve as piercing weapons—the sub-functions of the different parts or sections of the two

tools differ. For the composite spear, the functional organization involves at least three separate objects (a point, a shaft, and the materials linking the two), each with their own sub-functions (a working end, a handle, and a joint), one of which (the joint) we do not find in the simple spear. In contrast, the simple spear is a single object that has two sub-functional sections (the piercing point and the handle) located at different places on the same physical object, with no section serving a joint function. With these in hand, we can now identify three design sub-problems regarding the invention of the functional organisation of the hafted spear.

F.1—How should the working point and the shaft handle be organised so as to make a working composite spear?

There is an indefinite number of ways a stone point and a wooden shaft could be organised into a composite object, most of which would lead to a non-functioning result. Designing a functional economy—how the sub-functions of the components are organised in relation to one another—is what differentiates a working composite spear from a modern art sculpture: both have subcomponents fixed together, but only one provides a functioning end result.

Four design (sub-)sub-problems are relevant here. First, one had to decide what kind(s) of implement to use to serve as the working end. There was a range of existing stone tools providing various functions that could have been used at the time, such as hand-axes for cutting and piercing, blades and cleavers for cutting, scrapers to scrape, hammer stones to percuss, etc. Second, one had to decide how many working ends to include within the composite tool. Third and fourth, following Rots (2010, 22), one would have to decide the tool placement—i.e., the location of the physical connection between the working end and handle—and the orientation of the active part of the working end—i.e., the relative orientation of the working end relative to the handle[3]. Regarding tool placement, the working end could be set anywhere along the handle (lateral placement) or at either end (terminal placement). For orientation, the working end could have its working edge face any direction relative to the handle's axis.

Four necessary design choices were made here. For the hafted spear, a piercing point could have been used first, possibly a unifacial flake tool. It could have been otherwise, however.

[3] Rots (2010, 22) also includes tool direction, the general direction of movement when using the tool. Tool direction could be axial (along the handle) or transversal (perpendicular to the handle). However, for the present case study, tool direction itself is not a novelty as it is already found in the simple spear.

Cutting tools could have been tried out (e.g., cleavers, blades) or a completely new shape designed (e.g., inventing a point using existing shaping techniques). Second, only one implement was used to serve as a point, rather than many (e.g., the spear is not a trident). Regarding location and orientation, the working end was set at the tip of the handle (terminal placement), with its working edge pointing away parallel to the axis of the handle. These choices represent genuinely novel design solutions as they concern the relation between the working edge and handle components. For the simple spear, since it has no components, there is no question as to how many points to include, what the point should be made of, nor how the point and shaft are to be related to one another. While in hindsight these choices may appear trivial, they were nonetheless new for those dealing with component points and handles for the first time.

We also observe that there are many alternatives one could have chosen instead. A composite spear can have terminal hafting, with the point being inserted either at the very end of the handle, or along its side(s). For instance, harpoons have several of their working ends placed laterally but pointing mostly away from the tip of the shaft (Figure 3). This difference in functional organisations therefore implies that specific choices had to be made, but that they also were culturally transmitted, as different solutions could be locally adopted.

[FIGURE 3]

Finally, note that the functional organisation of the hafted spear differs from those of other hafted tools, such as the hafted knife, adze, and so on. For each of these alternatives hafted tools, different choices would have to be made (e.g., a hafted adze has the working end perpendicular to the axis of the handle). These are not trivial differences. Indeed, different design sub-problems pose themselves if one is to derive the functional organisation of a new hafted tool from an older one, such that inventing the functional organisation of one type of hafted tool does not necessarily provide solutions to the functional organisation of others[4].

[4] In fact, solving these sub-problems does not provide a general template for hafted tools, or a model of what a hafted tool, taken generally, should be like. Deriving such a general template may instead rely on either recognising a shared functional structure between existing hafted tools or could perhaps be derived through analogical reasoning. However, this question is better left to be addressed elsewhere.

F.2—How can the subcomponents working end and handle be linked with one another?

In order to put different material components together to form a hafted spear, one had to solve the problem of just how the working point could be associated with the handle so that the movements and forces applied to the handle would be transmitted to the working end. This problem is not one that inventing the simple spear needed to solve, as the working end is attached by default to the rest of the shaft. The solution to this problem was found through the functional novelty of *using an additional component* to serve the specific function of joining the working end to the handle. The invention of the join function is particularly significant as it adds to the repertoire of known functions not only a very versatile one, but also one with no precedent. Indeed, while one could reason through analogy how a hafted spear should be functionally organised by modifying the existing schema for the simple spear, the join function has no analogue in the design of hand-held simple tools (Figure 2). In contrast, both the working end and handle sub-functions already existed (on different sections of the simple spear), so these functions were not new.

Note that, conceptually at least, several alternative sub-functions could also have solved the problem of attaching two components together without requiring the invention of an intermediate join component. The shaft could be designed to snug the point in a way similar to how a hat grips one's head to remain in place. The point could have been fused to the shaft, as one could fuse a metal point to a metal shaft, therefore not requiring any additional components serving a specific join sub-function. Mixing different components can also lead these to attach to one another, as would ingredients linking together to make new composite substances, such as adhesives. Specifically, here, and in large part due to the physico-chemical constraints imposed by wood and stone (see next section), the solution to this design problem was to invent the join function and have a specialised component serving this function.

Of course, it is likely that binders and adhesives (see below) were independently invented to serve a join function before the hafted tool was itself invented. For instance, binders may have already been in use to attach animal skin to the human body, i.e., tied-up clothing may have existed prior to the invention of the hafted tool (Gilligan 2019). However, our analysis takes as a hypothetical precursor context one that does not include such uses and, doing so, we need to identify the invention of the join function as a novelty, i.e., as a relevant difference between our starting conditions and resulting invention, and one that was culturally transmitted (and still is

to this day). As for the study of any invention, just which design sub-problems constitute an invention process depends on the starting and end points chosen by the investigator.

F.3—How to deal with the fact that precursor components have non-detachable sections serving sub-functions conflicting with the new functional economy of the hafted tool into which they are integrated?

So far, the solutions to the two previous functional sub-problems F.1 and F.2 provide the general idea that one can use components to serve a join function, and what the general functional economy of a hafted spear is like. However, as we saw above, simple tools have non-detachable sections that serve different sub-functions. Unifacial flake tools have working edges, but they also have a 'handle' side to them, affording a precision grip, for instance (Barham 2013, 148; Shea 2016). The simple spear, if adapted into a shaft serving as a handle, also has a working end of its own, the piercing end. These sub-functional sections of the precursors—the 'handle' section of the simple point and the working edge of the simple spear—create conflicts if merely integrated within the overall functional organization of the composite spear (e.g., when hunting, one does not handle the composite spear by the bottom of the stone point). These sub-functions could not be preserved as they became components of the composite spear. The conflicting sub-functional sections of the simple tools had to be repurposed for these precursors to serve as components within their new functional economy.

As a solution, the functions of the conflicting sections of the point and shaft had to be rethought as serving a joining role. The point, instead of having a handle section, now has an end, serving the purpose of joining it to the haft. Inversely, the working edge of the single spear is repurposed to serve a join sub-function (Figure 2). These changes do not merely consist of inventing the joining function, as discussed in the previous sub-problem F.2. In the previous case, a new component was added with a specialised function. Here, it is the sub-sections of the precursor point and shaft that are repurposed. Neither join sub-functions existed prior to the invention of the hafted spear and, as we now turn to the tier of material implementation, there are many ways one can effectively implement this functional repurposing.

7. Inventing a material implementation

The combinatorial invention of the hafted spear required designing a novel functional organisation, i.e., the 'idea' of the hafted spear. One also had to design a hafting system capable of materially interfacing the different components together, such that the use-function

of the tool could be obtained. Finding whether and how the functional organisation of the hafted spear could be materially developed imposes a different range of problems that were both new and necessary to complete the invention of the composite technology. In fact, most of the problems in materially implementing the composite are those of designing a working hafting system. In this section, I focus on the invention of such material interface, leaving aside how the implementation of the working point and handle was solved (what materials the point and handle should be made of, whether they should be fire-treated, etc.).

I.1—Which materials and technologies can be used to implement the join function?

The idea of using additional components to serve a join function (F.2) necessarily imposes the (new) sub-problem of figuring out just which materials could be used to stick two separate objects together. To solve this problem, one had to parse, among all of the existing materials and technologies within their repertoire, which ones could effectively undertake this function. This meant not just finding a material that could attach itself to another, but one that could attach two different objects together. For instance, burnt foods can create char residue sticking to the branch supporting them over fire, but the leftover residues would not allow for additional materials to be fixed to the wooden stick.

Following Barham (2013, 184–93), in the context of the invention of the hafted spear, two kinds of implements have been found to serve this function: binders and adhesives. Binders can be used to wrap and attach two objects tightly by compressing them together such that the force applied to one component is transmitted to the other. Binders also ensure that the two bounded objects do not move independently from one another. Binders could be made of animal parts such as sinews, intestines and rawhide, or of plant fibre extracted from their bark, roots, stems, and even leaves, all made into cords. Adhesives glue together the surfaces of objects, not by compressing objects together but by having them cling to one another. Here again, the materials used came from animals and plants, such as processed cartilage, bones, and skin, or processed sap, resins and bark. Minerals such as iron oxide (red ochre) and tar could also be added. Barham also mentions fillers—materials used to fill unoccupied space within the haft—which either increase the compression potential of the binders or increase the surface for the adhesive to work on, or both. Here, tar and pitches could be used. While it is plausible that binders (and perhaps mastics and adhesives) were available before the invention of the hafted tool, for the present purpose I have assumed they were not (see Section 4 and 5).

In any case, should they have been available, the issue of choosing just which ones to use for assembling a working composite spear would remain.

I.2—How can the point and shaft be physically modified to ensure that their relevant sections can now serve a join sub-function?

As discussed above, the precursors of the point and handle were single object implements and had to be rethought as components of a composite whole, requiring repurposing some of their sections into serving a new join function (F.3). While this repurposing concerns how one thinks differently about the role of these sections in bringing about the broader functional economy of the composite spear, it was also necessary to invent a way to physically transform them so that they could serve their new purpose.

On the side of the shaft, the joining end would have to be transformed to permit the attachment of the point. Indeed, keeping the tip of the simple spear as a pointed end would create difficulties in finding a means to attach a point to it once it started to be used as a handle. Following Rots (2010), several kinds of haft could be devised, depending on where the point was to be joined to the shaft. Terminal hafts could be produced by cutting a slit at the end of the shaft (split haft), allowing one to slide the point between the extruding sides, or drilling the tip of the shaft (drilled socket) and inserting the point on its join end[5]. Lateral hafts could also be designed (see Figure 4). For the working point, several transformations could have been considered, such as the thinning or fluting of the joint sub-section or shaping it as a tang that would insert itself in the shaft's haft. Note that these are two different implementation problems but, because of their interdependence, for simplicity they are treated as one.

[FIGURE 4]

I.3—How can the different components of a composite tool be physically connected to ensure the use-function of the hafted tool?

Solving design problems I.1 and I.2 would provide means to fix a point to the end of a shaft and so satisfy the join sub-function (problems F.2 and F.3). It is a different problem altogether to find a working configuration of binders, adhesives and hafts such that they actually ensure that one can effectively use the composite spear as a piercing weapon. In other words, one had to

[5] Note that typically the

figure out just how to invent a configuration of different materials such that the manipulation of the shaft component transmits the mechanical forces applied to the point and permits the overall use-function of the tool.

There are several ways to haft a working spear, and the specific configuration one uses will depend heavily on the range of materials one uses (problem I.1), the kinds of physical modifications one makes to the point and shaft components (problem I.2), but also on the range of forces and stresses that the use-function will impart on the composite tool. Producing a working composite spear demands arranging the adhesives, binders, and hafts together so that the forces applied to the handle are adequately transmitted to the working end while at the same time allowing the composite to resist the various mechanical stresses involved in its use.

As hafted tools are meant to be used to do physical work—piercing, cutting, scraping, etc.—their use involves a range of different forces and mechanical stresses that act on the composite and its components, and between the components (Barham 2013, 178–84). Manipulating a hafted spear involves the force used on the shaft being transmitted to the point I parallel to the shaft's axis. A hafted spear is intended to pierce through some material by perforating the surface of the material on the working edge of the point. As the working edge of the spear hits the animal, the counterforce imparted on the point's working edge travels backwards towards the handle, compressing the haft and fixing the point to the handle. Moreover, as the working edge of the point may get stuck in the animal, the user may need to release the spear through motions that will act as torsion between the animal and the point, and therefore between the point and the haft, and the haft and handle. These three types of stresses (axial loading with compressive stress, tensile stress, and torsion) will demand a haft resilient to these so that the spear can be used for more than one hit (or at least one hit). These problems are new to the invention of the hafted spear because they concern the physical relations between the component parts—and different solutions will need to be found for other hafted tools undergoing different kinds of physical stress.

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These three sub-problems of implementation were new to the invention of the hafted spear, as they involve finding material solutions that ensured the components would physically work together to form a functional weapon. Without solving these complex issues, there could be no usable composite spear. While there exist many potential configurations of binders, adhesives and hafts (including the materials used to make them), these are far from being obvious problems to solve. It would take a very lucky individual, or more likely a lot of trial-and-error

experimentation, to figure out just how to solve the sub-problems posed by the forces and stresses involved in the use of the composite spear, including just how to find an acceptable trade-off between these different demands. In fact, this balancing is complex enough to constitute a combinatorial invention of its own, with the specific choice of precursor materials and their transformation to be put together into a working joint component. It is therefore necessary for the solutions found to these design sub-problems be passed along culturally, and the archaeological record does display variation in the sub-solutions adopted by different populations, as exemplified in the works of Rots (2010) and Barham (2013).

8. Inventing an assembly procedure

The two previous tiers identify the range of design problems that need to be solved for producing a working hafted spear. These concern the characteristics of the composite invention as an end product, i.e., as a functioning weapon. However, they do not tell us which means were used to fabricate the tool itself. Of course, to solve the material implementation problem, one must have been able to effectively assemble the different materials and components to make the tool, but we have not yet examined the procedures used for doing so, and whether there are novel design problems related to those procedures. Here, I identify two such design sub-problems.

A.1 How does one assemble a hafted spear?

According to Kuhn (2020, 45–48), there are two main ways in which one can organise the production of a hafted spear, which he calls Mode A and Mode B[6]. Mode A consists of the general strategy of adapting the shaft that will serve as a handle to the specific morphological properties of the stone implement used as a working end. This would involve first shaping a stone point and then choosing a shaft and customising its hafting end to the specific shape of the point. Mode B consists of the reverse: first, a handle is selected and from it, one would shape the stone point to fit the tip of the shaft. Kuhn proposes that the choice of either mode will likely reflect the level of investment, durability, and preciousness of the materials used. For points that are cheap to produce and break easily, Mode B will be preferred, as it allows the reuse of the handle while replacing the expendable points. Inversely, if the point is costly to

[6] Kuhn describe a third mode, Mode C. Since this mode concerns specifically the assembly of hafted tools using microliths, it will not be discussed in the present context.

produce or depends on rare materials, the shaft instead may be adapted to the point and the point reused when requiring a new handle (Mode A).

Many steps within these assembly procedures were already known and would not require inventing. The production of the point and of the haft components may be borrowed techniques used for producing their single object counterparts. However, inventing any one specific procedure would require combining existing techniques together to form the more complex assembly procedure of the hafted spear. In this sense, the new assembly procedures would themselves be combinatorial inventions in their own right (see Charbonneau [2016] for a discussion of combining and recombining technical modules). In turn, such combinatorial sub-inventions could give rise to potentially new sub-problems—e.g., when should one start preparing adhesives relative to the production of the point and shaft? Should the procedure be done by a single individual or should there be a division of labour? For reasons of space, I leave such questions open.

Each of the two modes would have to be new to the invention of the hafted spear as they concern the specific procedures to join a component working end to the handle. Making a simple spear (Haidle 2009; 2010) does not involve the problem of deciding which component of the spear should be adapted to the other simply because it has no components. While we do not know which mode was first used, together these two modes are disjunctively necessary, as without an assembly procedure, no composite spear could be produced at all. Moreover, the assembly procedures must have been transmitted along with the solutions to the functional organisation and material implementation problems for the hafted spear to become a technological tradition. As documented by Barham (2013) and Kuhn (2020), we have empirical evidence of both modes A and B having been used in different communities at different times.

A.2 How does one produce the joint components (binders, adhesives, haft)?

There are several ways any binder or adhesive can be made, and different materials could and have been employed (see Barham [2013] for discussion and references). Most of these materials, however, are not naturally available such that they could simply be harvested and applied to the point and shaft. Binders have to be processed in some way or another (e.g., by making fibres into cordage, soaking then drying sinew and intestines), and so do adhesives (e.g., pyrotechnically extracting tar from bark). Examining any specific variant method goes beyond the scope of this analysis, but it is safe to say that any solution would have required

solving complex design problems regarding the procedures, given how complex such techniques had to be (Wadley et al. 2009; Wadley 2010). Given the complexity of this issue and its existing literature, it will not be further discussed here. For more detailed discussion, see Barham (2013, 185–93) and Rots (2010, 16–19) and references therein.

9. Discussion

We can now summarise the results of this preliminary analysis. Inventing the composite spear is a complex process that involved solving multiple design sub-problems, several of which in turn required solving design sub-problems of their own. For reasons of space, the analysis has only scratched the surface, focusing on those problems that are best understood as the first layer of design sub-problems involved in the combinatorial invention of the hafted spear. While the identification of design problems related to the functional organisation of the hafted spear appears close to complete, a deeper analysis is required for the problems of material implementation and procedural assembly. For instance, further work is required to identify the design sub-problems involved in inventing binders and adhesives and the sub-problems related to balancing together specific forms of hafting, binders, and adhesives. Indeed, the choice of specific materials can create sub-problems of their own, different from those encountered when using other materials. Problems related to finding and pre-processing the materials, especially those used to form the new joint component, also remain to be examined. The analysis of the assembly procedures is particularly preliminary and further work—such as adopting a finer-grained analysis of technical actions and decision components (e.g., see Charbonneau [2016]; Haidle [2009]; Stout [2011])—promise more sub-problems involved in figuring out the interface between the different steps deployed to build a usable hafted spear. Moreover, the examination of the phylogenetic traces of the variant solutions to the sub-problems of each tier has been mostly cursory. A finer, more systematic examination is required to fully illustrate the range of alternatives that have been devised and to confirm their cultural nature. Importantly, this means that the components (be they functional dimensions, material interfaces and changes, or new procedures) should be understood as ‘traits’ forming lineages of their own, even if entangled in the complex tradition of the hafted tool. This requires a finer analysis of lineages than merely tracking how the broader artifacts have changed and have been combined with one another. Tracking how these (sub-) components evolved in their own way across time and space will provide a finer, more instructive understanding of invention as both a problem-solving and a lineage-making process. Nevertheless, we can already see how the proposed framework can produce insights into the combinatorial invention process,

and how understanding invention as a problem-solving process producing novel lineages allows us to open the black box of combinatorial invention, and of invention more broadly.

The present analysis is also preliminary as it has only focused on determining the range of sub-problems involved in inventing the hafted spear. A more detailed analysis should also address the problem-solving processes and activities involved in figuring out solutions to those problems. Evolutionary models and some cultural evolutionists understand invention first and foremost as a chance process requiring little if any intelligence (Boyd et al. 2011; Henrich 2016), depicting invention—and therefore combinatorial invention—as mostly a serendipitous process (Muthukrishna and Henrich 2016). While the present analysis does not reject this understanding, it does underline the inadequacy of the oversimplified image of combinatorial invention as a single serendipitous event rather than one consisting of a prolonged process of solving design problems and sub-design problems. Pre-*sapiens* hominins did not simply stumble by chance into assembling points, shafts, binders and adhesives into a fully formed and functional hafted spear, if only because of the astronomical space of potential dysfunctional combinations of all these precursors. Instead, the combinatorial invention of complex technologies is more akin to a prolonged, intergenerational process of problem-solving, involving multiple inventions and sub-inventions, each of which can, in turn, be combinatorial in their own right. The present analysis has shown that if combinatorial invention is a chance process, it is one that would comprise a complex series of serendipitous accidents.

Alternatively, combinatorial invention could be driven by a range of head-scratching problem-solving activities (Wragg Sykes 2015). Cognitive mechanisms such as technical reasoning (Osiurak and Reynaud 2020) and planning (Ambrose 2001; 2010; Barham 2013; Wynn and Coolidge 2011) could have played an important role in solving many of the problems within the implementation and assembly tiers. The capacity for compositional thinking and manipulating hierarchical organisations (Greenfield 1991; Stout 2011; 2021), together with analogical reasoning (de Beaune 2004; de Beaune et al. 2017), are also likely candidates to understand how combinatorial inventions come about. Cognitive mechanisms could also have made some aspects of the invention of the hafted spear more difficult. We know that Modern humans are prone to functional fixedness in problem-solving, i.e., a difficulty in repurposing objects with known functions (Duncker 1945), and that this is also found in populations with smaller technological repertoires (German and Barrett 2005). It is an open question whether (and if so, which) pre-*sapiens* hominins were also prone to such bias, and whether it influenced the solving of sub-problems within the various tiers, especially the functional one (Section 6). While cognitive archaeologists have focused mostly on elucidating the cognitive underpinnings of the

procedures to fabricate and use technologies (e.g., Haidle 2009; Lombard and Haidle 2012), in a similar spirit the framework proposed here can provide the grounds to assess which cognitive mechanisms could have been involved in the process of invention by providing a structured decomposition of the design problems that had to be solved.

Whether such inventions were mere affairs of chance, hard thinking, or a mix of the two, any approach to explaining an invention process will require organising the different sub-problems into a gradual evolutionary sequence where each step satisfies the immediate goals and needs of those who made the inventions. By approaching invention as both an individual-level process of problem-solving and as a population-level process of lineage formation, the framework developed here provides the analytical grounds to chronologically (or serially) reconstruct the extended process of combinatorial invention of complex technologies into a series of sub-inventions, and therefore to compare the plausibility of specific evolutionary scenarios.

The analysis also reveals that the combinatorial invention of complex technologies frequently entails more than just merging existing technologies to make new ones. Instead, it requires developing novel interfaces between these components, interfaces that themselves become distinct technological traditions. In the case of hafted tools, the invention of hafting systems represents more than just a means of joining a working end to a handle; these solutions evolved into variant technological lineages with their own evolutionary history of cumulative improvements, leading to parallel evolutionary trajectories. This challenges the assumption of evolutionary models that combinatorial invention can be reduced to a simple mapping between precursor inputs and a composite output. The present analysis instead suggests that combinatorial invention should be understood not merely as a process that produces composite technologies, but as one that generates multiple concurrent technological lineages through the solving of multiple design sub-problems. Whether this conclusion also challenges the results of evolutionary modelling or would simply make them more grounded remains a question open to further investigation.

10. Conclusion

Here, I developed a framework for studying invention, and combinatorial invention specifically, by recognising its dual nature as a design problem-solving process generating novel technological lineages. By identifying genuine design sub-problems—those requiring solutions that were new, necessary and culturally transmitted—we can reconstruct aspects of the combinatorial invention process through the phylogenetic traces these solutions left in the

record. These traces manifest as multiple technological traditions that evolved alongside the main composite technology, each representing solutions to specific design sub-problems encountered during the invention process. This approach transforms what might appear as an unknowable creative event (Kuhn and Stiner 1998; Kuhn 2020) into a partially reconstructible process through the careful analysis of the constitutive and supportive technological lineages it spawned. Unlike existing evolutionary models that treat combinatorial invention as a process of randomly matching components in a lock-and-key fashion, this framework reveals the complex problem-solving process involved in making precursor technologies work together through carefully designed interfaces.

11. References

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Figures

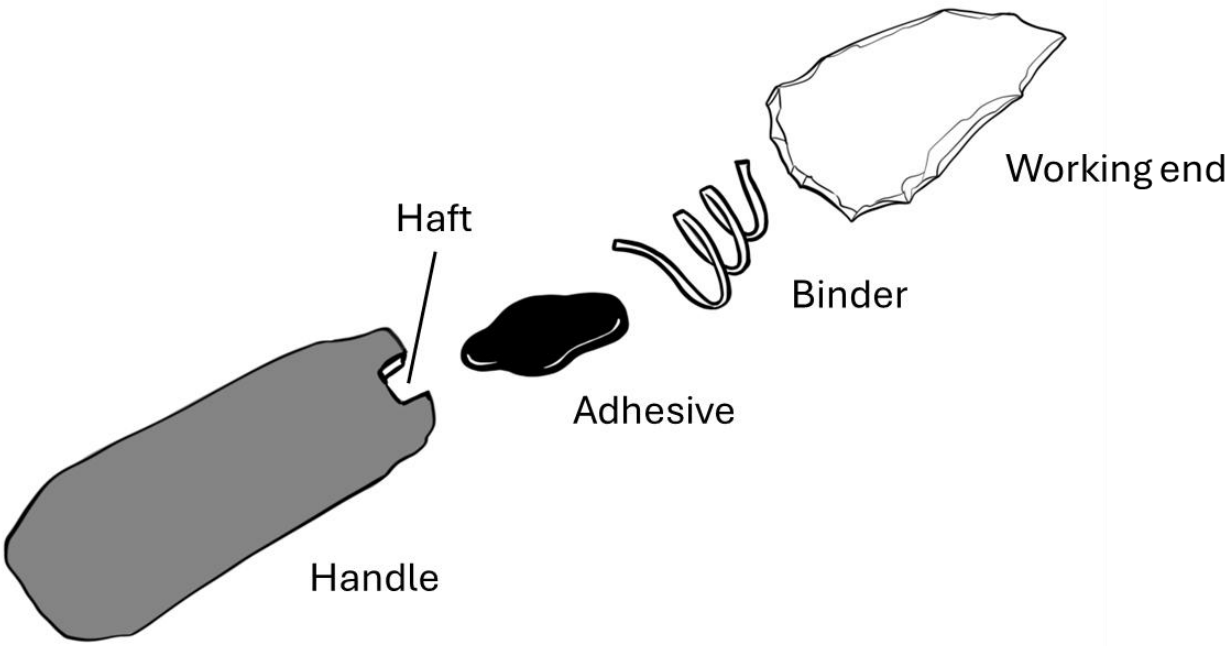


Figure 1 – Components of a hafted tool (here a hafted knife), organised by function. Adapted from Barham (2013, p.2) [Permission still required] (Illus. Rania El Maloumi)

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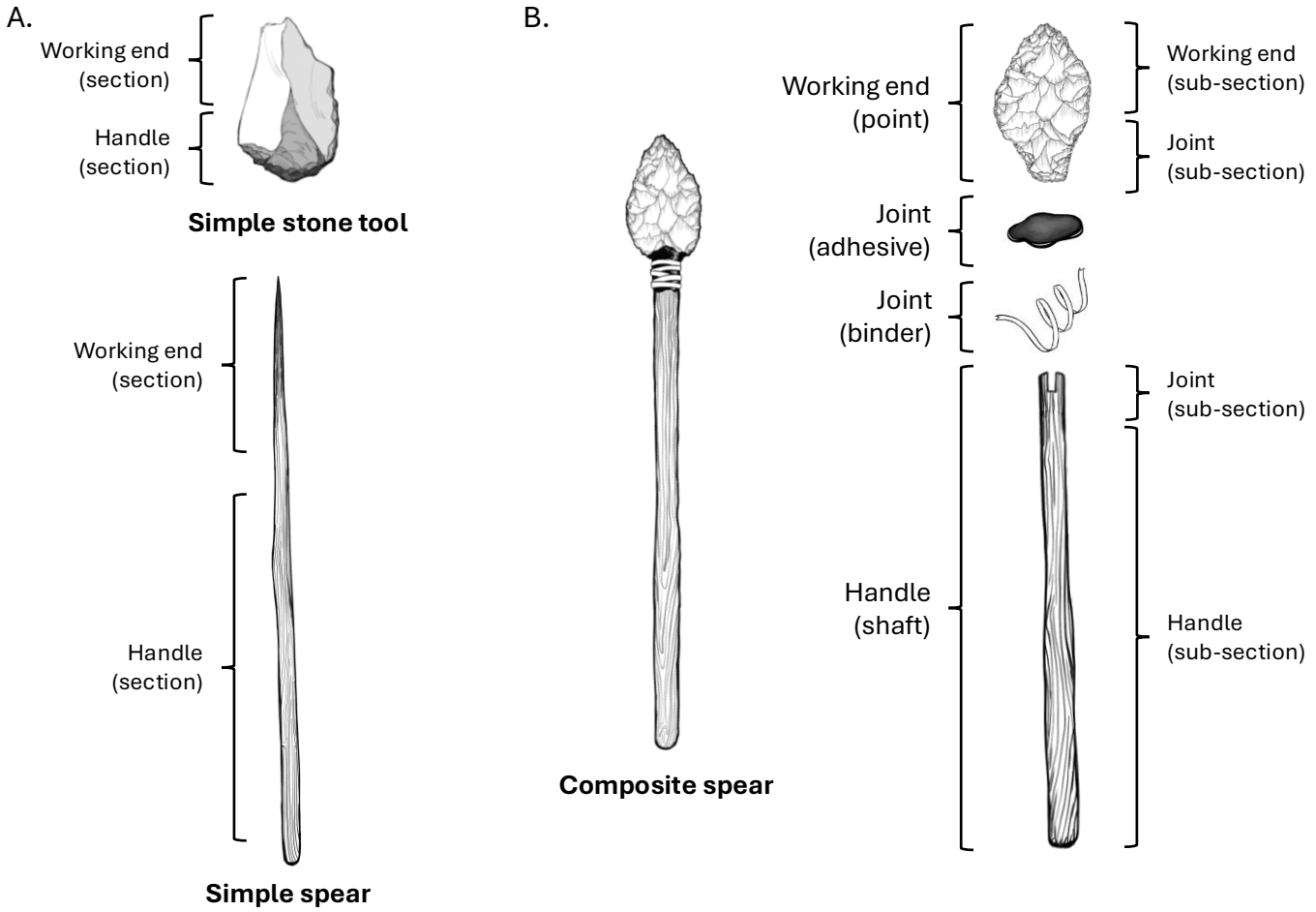


Figure 2 – Part-whole and functional decomposition of the hand-held stone tool (unifacial flake), simple pear, and composite hafted spear. A. Simple stone tool (unifacial flake) and simple spear each have two non-detachable functional sections, the working end (acting as a cutting or piercing side) and a handle. B. The composite spear is composed of several parts, each of which serves a specific purpose. The point serves as a working end, and has two non-detachable sub-sections, one serving a working end sub-function and the other a joint sub-function. The shaft serves as a handle and has two non-detachable sub-sections, one serving as a joint sub-function (as haft) and a handle sub-function. Binders and adhesives serve as detachable parts serving a joint function, in tandem with the join sub-sections of the point and handle. (Illus. Rania El Maloumi)

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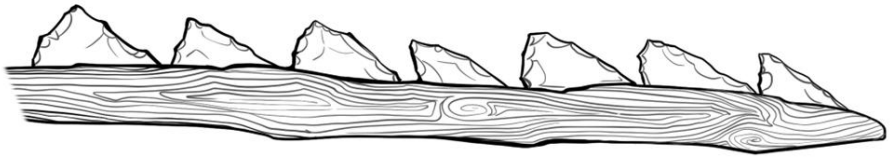


Figure 3 – Tip of a harpoon, with multiple working ends, placed along a lateral haft, with their working edge (cutting) pointing obliquely away from the axis of the shaft. (Illus. Rania El Maloumi)

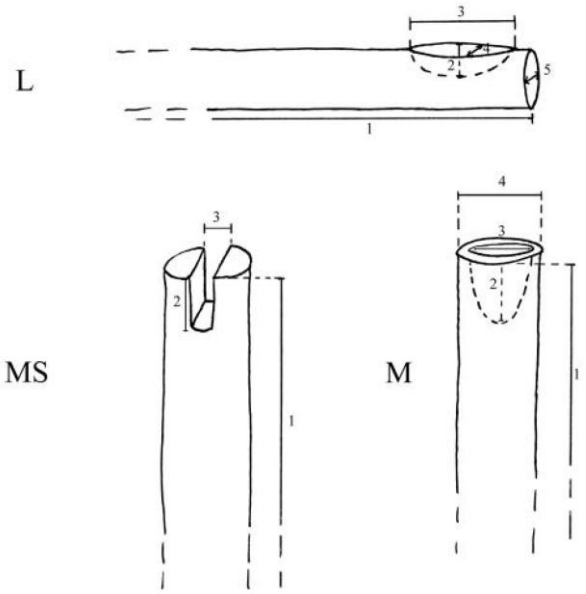


Figure 4 – Three types of hafts. L = Lateral haft. M = drilled socket. MS = split haft. Adapted from (Rots 2010, 22). [Permission still required]